

PAPER

# Determining Planck's constant with LEDs—what could possibly go wrong?

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# Determining Planck's constant with LEDs—what could possibly go wrong?

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## Abstract

Light emitting diodes have been used to determine Planck's constant in introductory physics laboratories. One common method relies on the energy of the light emitted by the LED and its relation to the energy gap in the solid of which the diode is composed. However, there could be a problem with the data that are collected for this experiment. For some LEDs the energy of the light emitted from the LED can be quite different from the gap energy. If these LEDs are used for the experiment, the results will give different results for Planck's constant.

## 1. Introduction

Light emitting diodes have been used to determine Planck's constant in introductory physics laboratories. One common method relies on the energy of the light emitted by the LED and its relation to the energy gap in the solid of which the diode is composed. However, there could be a problem with the data that are collected for this experiment. For some LEDs the energy of the light emitted from the LED can be quite different from the gap energy. If these LEDs are used for the experiment, the results will give different results for Planck's constant.

As light emitting diodes have become ubiquitous light sources, physics instructors have used them in a variety of ways for educational purposes [1–5]. Among the many experiments that students can do is the determination of Planck's constant using LEDs. Two different methods have been discussed in the literature. One approach uses the standard photocell equipment with LEDs of various colours for the light sources [6]. The

other relies on the energy band and energy gap structure of semiconductors to obtain an estimate of Planck's constant. We will discuss this latter method and a difficulty that can be introduced by some LEDs which are commonly called ultrabright.

Perhaps the first article to discuss the LED energy structure and a method of using that structure to determine Planck's constant appeared in 1974 [7, 8]. Over many years, descriptions of this experiment have appeared in journal articles [9, 10] and on the Web. A search for 'Planck constant LED' will provide links to many documents, many of which are lab write-ups created for students, and some YouTube videos.

## 2. Background

A simple semiconductor, such as an LED, consists of two materials, each of which has energy bands and an energy gap. When we apply a voltage to an LED, we cause electrons to move from one material to the other and then to emit light

by a transition from the conduction band to the valence band. The voltage at which light is first emitted is called the threshold voltage. Thus, the energy gap of the semiconductor in an LED is related to the threshold voltage of the LED and to the energy of light emitted by it. The relation is:

$$E_{\text{gap}} = E_{\text{light}} = eV_{\text{threshold}} \quad (1)$$

where  $E_{\text{gap}}$  is the value of the energy gap and  $E_{\text{light}}$  is the energy of the photon emitted by the LED. The energy of a photon ( $E_{\text{light}}$ ) is related to its frequency ( $f$ ) by the Planck–Einstein equation:

$$E_{\text{light}} = hf_{\text{light}} \quad (2)$$

where  $h$  is Planck's constant. Thus, Planck's constant can be determined by measuring the threshold voltage and the frequency of the light being emitted by an LED. Holcomb has pointed out that this analysis is only an approximation [11].

### 3. Student experiment

In typical experiments students measure the threshold voltage of several LEDs and plot this voltage versus the frequency (or wavelength) of the emitted light. Planck's constant is then determined from the slope of the line fitting the frequency versus voltage data. The primary difficulty for students is determining when the LED first begins emitting light. The simplest method is to have students judge when they first see the light from the LEDs. This method has some obvious drawbacks but results in an estimate of  $h$  that is the correct order of magnitude. Alternatively, students can measure both the current and voltage in the LED circuit. When current begins to flow, the LED also begins to emit light, even though the light intensity can be too weak to see.

Thus, this quantitative approach is much more accurate than student judgement of the turning on of the LED. A third approach is to use some type of light detector in conjunction with the voltage measurement. Voltage versus light intensity measurements curves can be extrapolated to obtain the threshold voltage at which light is first emitted. This approach has been successful using mobile phones as the light detector [10].

Each of these methods provide a pedagogically sound way for students to see the connection between the energy bands and gaps in solids and the energy of light emitted by photons from

LEDs. Accuracy can range from order of magnitude to better than 10%.

### 4. The surprise

Each of the authors of this paper independently discovered a flaw in this process when using modern, commercially available LEDs. That flaw is represented in figure 1. The surprise is a 'bright green' LED with a frequency of  $5.71 \times 10^{14}$  Hz. As seen in figure 1, the value for the threshold voltage lies significantly off the line for the other LEDs. This anomaly is even more apparent in the videos at <https://youtu.be/uy-xTadGobY> and <https://youtu.be/r7XFX1rY8fA>. In the videos the voltage across all of the LEDs is increased simultaneously. If the photons are directly emitted from the semi-conductor, the LEDs will turn on in order of photon energy as the voltage is increased.

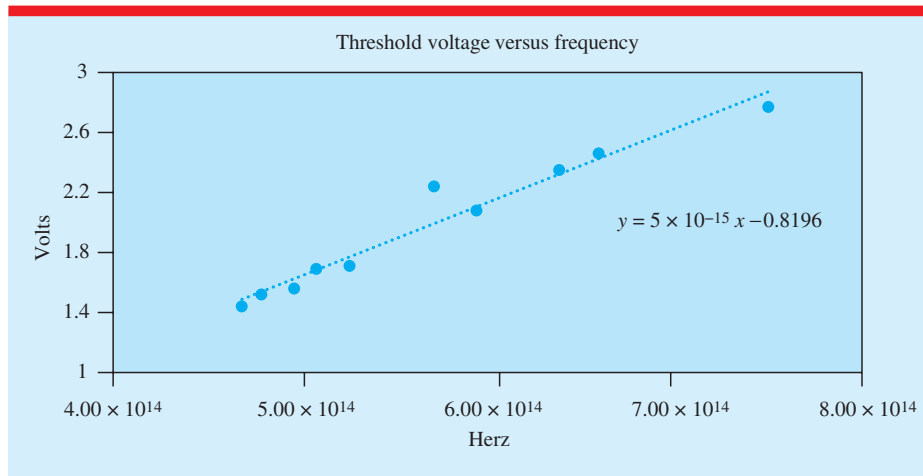
Figure 2 shows four still photos from the video. In the first two images the LEDs are turning on at appropriate voltages and in order of photon energy. However in the third image (2.62 V) the turquoise LED is on but the bright green one just above it is still off. Finally as the blue and violet LEDs come on (3.09 V), the bright green one does also. Something is different about this bright green LED.

### 5. The process

The other LED which is not shown in any of the photographs is the white LED. Its behaviour is very similar to the bright green one. Even though white light is a combination of all colours, the white LED does not come on as red when the red LED turns on, then add green at the threshold voltage for a green LED, and so on. Instead, the white LED is off until the voltage reaches that of a blue or violet LED (slightly more than 3 V). At that voltage, the LED emits white light

Fluorescence is the process behind both the white and ultrabright green LEDs. The LED inside these devices emits blue light which is then absorbed by a fluorescent material. The material is constructed to convert the blue light into the desired frequency or frequencies. This process is called 'down conversion' because it changes relatively high energy (frequency) blue light into lower energy (frequency) light. For the white LED some of the blue light is down-converted to a broad spectrum of colours

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**Figure 1.** A graph of threshold voltage versus frequency for a typical set of LEDs ranging from red ( $4.69 \times 10^{14}$  Hz) to violet ( $7.5 \times 10^{14}$  Hz). The voltages are measured as described in the text, while the frequencies are those given by the manufacturer. The dotted line is a linear fit to these data.



**Figure 2.** A series of still photos from the video showing the LED behaviour as the voltage across them is increased. At 2.62 V the bright green LED which emits light at  $5.52 \times 10^{14}$  Hz is still off even though the turquoise one which is directly below it and emits light at  $5.94 \times 10^{14}$  Hz is already on. (The apparatus which contains the LEDs and shown in the photos is sold in the US by Arbor Scientific. It is *not* designed for have a variable voltage applied to it.)

peaking at yellow. That colour is mixed with the remaining blue light to achieve white light. For the ultrabright green LED all of the blue light is down-converted to green. This process is illustrated in figure 3. Also see [12] which is written for the entertainment industry and thus focuses on colours of white LEDs.

### 6. Why?

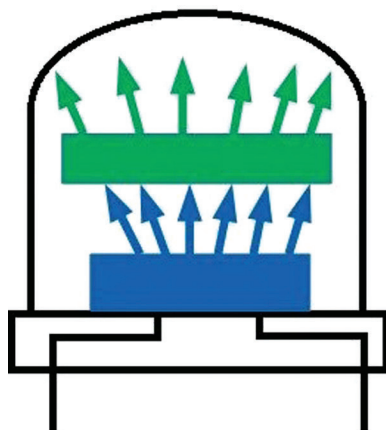
In the manufacturing of semiconductors such as LEDs the sizes of the energy bands and energy gaps can be controlled by the selection of elements and their relative amounts that are used to make them. For example, a common material for red LEDs is aluminium gallium arsenide while yellow and green LEDs are made of gallium phosphide. The ability to achieve a bright output

from a material that has a band gap in the green part of the spectrum is rather limited. These so-called intrinsic green LEDs are manufactured but tend to be limited in light intensity.

Down-conversion LEDs emitting green light, on the other hand, are easily manufactured and readily available. They generally use indium gallium nitride to produce blue light that is down-converted to green [13]. The bright green light produced in this way is good for any purpose except measuring Planck's constant. However, for sufficiently advanced students, this anomaly can provide an interesting experimental challenge.

### 7. Conclusions

Using the threshold voltage of light emitting diodes to measure Planck's constant is an instructive



**Figure 3.** In the ultrabright green LED a blue LED emits light. That light is down-converted to green by a fluorescent material.

experiment for students. It can help students make connections with quantum properties of solids and see how we can determine these properties even though we cannot see them. It can further help students see the connections between contemporary physics and everyday modern lighting devices. However, as the LED industry continues to develop new and brighter quantum lighting devices, we must be careful in our selection of the LEDs for this experiment. Otherwise, when the students find a bump in their curve, they just might be correct.

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**Ian Bearden** has been on the faculty of the Niels Bohr Institute, University of Copenhagen since 2003. He is the chair of the Danish Physical Society's Section for Education and Outreach, and the Director of NBI's Ungdomslaboratoriet (YouthLab). Ian is a member of the ALICE

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